

Roll No



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech. IIIrd Sem

Mid Semester Examinations, Sep 2026

Course Name: **Fluid Mechanics**

Date: 25-Sep-2026 (Friday)

Duration: 1:30 Hours

Course Code: **ME202**

Time: 10:00 AM - 11:30 AM

Max. Marks: 50

ANSWER ALL QUESTIONS

All dimensions are in cm, if not otherwise mentioned.

No clarification needed for any question. If you think that there is a something missing, please assume.

1. Derive 3-dimensional continuity equation in cylindrical coordinates from cartesian coordinates, for incompressible Newtonian flow. Take, $r^2 = x^2 + y^2$, such that $x = r \cos \theta$, $y = r \sin \theta$, $\theta = \tan^{-1} y/x$,

$$\frac{\partial x}{\partial t} = V_x, \quad r \frac{\partial \theta}{\partial t} = V_\theta,$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial x}, \quad \frac{\partial}{\partial y} = \frac{\partial}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial y},$$

Show,

$$\frac{1}{r} \frac{\partial(rV_r)}{\partial r} + \frac{1}{r} \frac{\partial(V_\theta)}{\partial \theta} + \frac{\partial(V_z)}{\partial z} = 0$$

[10]

2. Find the pressure at point A and B of the pitot tube (Fig. 1). What are these pressure called, and what is the value of dynamic pressure and the mass flow rate in the pipe (radius 5 cm). $SG_{manometer} = 13.6$ and $SG_{flow} = 0.85$

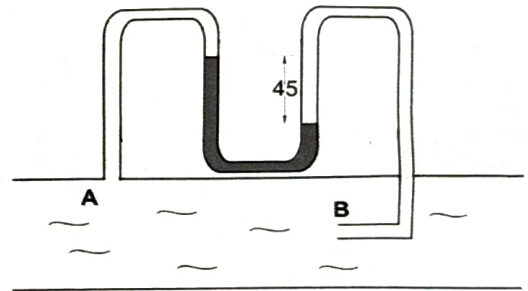


Figure 1

[5]

3. Show that a forced vortex flow in a cylindrical cup makes a parabolic free surface. Use Bernoulli's Equation if necessary.

OR

Show that the pressure at the outlet of radial flow is less than that of a inlet flow. Draw the radial change of pressure and velocity graph across the outlet to inlet.

[5]

4. Find the exhaust velocity of the an aeroplane engine fitted with blower giving 600 m head work and outlet pressure 60 kPa. Use Bernoulli's equation terms for per unit specific weight. [5]
5. Explain Area of Vena-Contracta and Coefficients of hydraulic discharge, velocity and contraction.

AND

Find the speed and rate of discharge at the outlet tap of a tank of diameter 15 m, and length 30 m, contains petrol ($\rho = 750$), is placed horizontally at 5th floor of a building. Assume each floor height is 10 ft. The outlet tap has an opening of 1 inch diameter and placed at the bottom of the tank. Under normal conditions, what will be actual velocity at the tap outlet? Given, $C_v = 1.1$, and $C_d = 0.8$

[5+5]

6. Discuss stability of a floating body and submerged body.

AND

For a floating can in water as shown in Fig. 2. Find the SG and weight of the can.

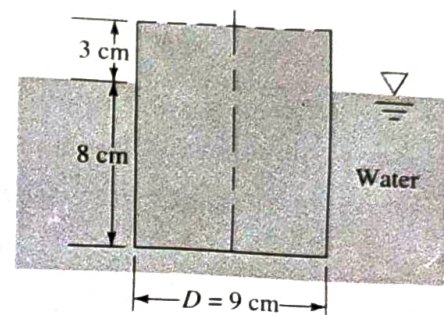


Figure 2

[5+5]

7. A curved gate AB holding water as shown in Fig. 3 experiences a resultant force **R**. Find magnitude and direction of **R**.

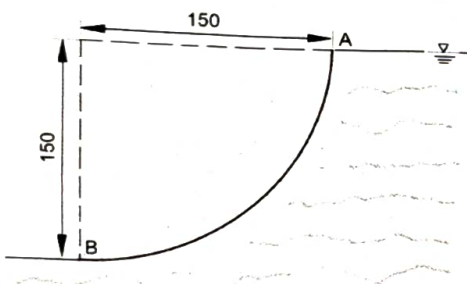


Figure 3

[5]

*** ALL THE BEST ***